Name: R.Jagan

Reg No: 192321102

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Machine Learning – C02 – AT1

Question 1
A Genetic Algorithm is applied to maximize $f(x)=x^2$ using the initial population {2,4,6,8}.

Individual (x)	Fitness $f(x)=x^2$	Remarks
2	4	Low fitness
4	16	Moderate fitness
6	36	High fitness
8	64	Best fitness

Step 1: Fitness Evaluation
The fitness of each chromosome is obtained by substituting x into $f(x)=x^2$.

Fitness values:
2→4
4→16
6→36
8→64

Step 2: Selection
Since the objective is maximization, the chromosomes with the highest fitness are selected.
Selected parents: x = 8 (fitness = 64) and x = 6 (fitness = 36).

Step 3: Crossover (Exchange of Least Significant Bit)

Binary representation:
8 = 1000
6 = 0110

Least Significant Bits (LSB):
1000 → LSB = 0
0110 → LSB = 0

After exchanging the LSBs:

1000 → 1000 = 8
0110 → 0110 = 6

Offspring generated:
Offspring 1 = 8
Offspring 2 = 6

Since both LSBs are identical, the offspring remain unchanged. In practice, crossover may or may not create new solutions depending on the selected crossover point and parent chromosomes.

Role of Mutation

Mutation randomly changes one or more bits of a chromosome with a very small probability.
Example: 1000 (8) → 1001 (9) by flipping the last bit.

Importance:
• Introduces new genetic material into the population.
• Prevents premature convergence to local optima.
• Maintains population diversity.
• Improves exploration of the search space.
• Increases the probability of finding the global optimum.

Question 2

A perceptron has:
Weight $w_1 = 0.5$
Weight $w_2 = 0.3$
Bias $b = -0.4$
Inputs:
Advertisement exposure $x_1 = 2$
Income level $x_2 = 1$

Step 1: Net Input Calculation

$\text{Net} = w_1x_1 + w_2x_2 + b$

$= (0.5 \times 2) + (0.3 \times 1) - 0.4$

$= 1.0 + 0.3 - 0.4$

$= 0.9$

Step 2: Step Activation Function

Step function:
Output = 1, if Net ≥ 0
Output = 0, if Net < 0

Since Net = $0.9 \geq 0$,
Output = 1

Final Decision

The perceptron classifies the customer as:
Purchase = YES (Output = 1)

Conclusion:
The computed net input is 0.9. After applying the step activation function, the output is 1, indicating that the customer is predicted to purchase the product.